

HW 6

To the grader:

I'm dropping this homework, I did some of the problems but no where near all.

Townsend 5.14 (5 pts)

The first part of this problem follows the procedure outlined in Problem 3.5. The operator

$$\hat{R}(\theta\mathbf{i}) = e^{-i\hat{S}_x\theta/\hbar}$$

rotates spin states by an angle θ counterclockwise about the x axis.

- (a) Show that this rotation operator can be expressed in the form

$$\hat{R}(\theta\mathbf{i}) = \cos \frac{\theta}{2} - \frac{2i}{\hbar} \hat{S}_x \sin \frac{\theta}{2}$$

Suggestion: Use the states $|+\mathbf{z}\rangle$ and $|-\mathbf{z}\rangle$ as a basis. Express the operator $\hat{R}(\theta\mathbf{i})$ in matrix form by expanding $\hat{R}$ in a Taylor series. Examine the explicit form for the matrices representing $\hat{S}_x^2$, $\hat{S}_x^3$, and so on.

- (b) Use matrix mechanics to determine which of the results $|\Psi_{12}^{(+)}\rangle$ or $|\Phi_{12}^{(-)}\rangle$ of Alice's Bell-state measurement yields a state for Bob's particle that is rotated into the state $|\psi\rangle$ by the operator $\hat{R}(\pi\mathbf{i})$.

A

TODO

b

Townsend 5.21 (5 pts)

The density matrix for an ensemble of spin- $\frac{1}{2}$ particles in the S_z basis is

$$\hat{\rho} \xrightarrow{S_z \text{ basis}} \begin{pmatrix} \frac{1}{4} & n \\ n^* & p \end{pmatrix}$$

- (a) What value must p have? Why?
- (b) What value(s) must n have for the density matrix to represent a pure state?
- (c) What pure state is represented when n takes its maximum possible real value? Express your answer in terms of the state $|+\mathbf{n}\rangle$ given in Problem 5.19.

$$\langle S_y \rangle = \text{tr}(\hat{S}_y \hat{\rho})$$

$$p_{ij} = \langle i|\psi\rangle \langle \psi|j\rangle$$

A

$\frac{3}{4}$ - the trace has to add to 1 for any ensemble.

B

We know that for a pure state, $\hat{\rho}^2 = \hat{\rho}$.

$$\begin{aligned} \rho = \rho\rho &= \begin{pmatrix} \frac{1}{4} & n \\ n^* & \frac{3}{4} \end{pmatrix} \begin{pmatrix} \frac{1}{4} & n \\ n^* & \frac{3}{4} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{16} + n^*n & n \\ n^* & n^*n + \frac{9}{16} \end{pmatrix} \end{aligned}$$

This tells us that $n^*n = \frac{3}{16}$

with any complex phase, so

$$n = \frac{\sqrt{3}}{4} e^i$$

The density matrix is now

$$\begin{bmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} e^i \\ \frac{\sqrt{3}}{4} e^{-i} & \frac{3}{4} \end{bmatrix}$$

C

The maximum possible real value is $n = \frac{\sqrt{3}}{4}$

We know that a general state

$$|n\rangle = \cos \frac{\theta}{2} |+\rangle + e^{i\phi} \sin \frac{\theta}{2} |-\rangle$$

We can compare this with

$$\begin{bmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} e^i \\ \frac{\sqrt{3}}{4} e^{-i} & \frac{3}{4} \end{bmatrix}$$

The diagonal terms of the density matrix with $|n\rangle$ are $|+\rangle\langle+|$, so we just have the elements squared.

$$\cos^2 \frac{\theta}{2} = \frac{1}{4}$$

$$\cos \frac{\theta}{2} = \frac{1}{2}$$

$$\sin^2 \frac{\theta}{2} = \frac{3}{4}$$

$$\sin \frac{\theta}{2} = \frac{\sqrt{3}}{2}$$

because we have the largest real value, $\phi = 0$, so we have the state

$$|n\rangle = \frac{1}{2} |+\rangle + \frac{\sqrt{3}}{2} |-\rangle$$

Townsend 5.23 (5 pts)

An attempt to perform a Bell-state measurement on two photons produces a mixed state, one in which the two photons are in the entangled state

$$\frac{1}{\sqrt{2}} |x, x\rangle + \frac{1}{\sqrt{2}} |y, y\rangle$$

with probability p and with probability $(1-p)/2$ in each of the states $|x, x\rangle$ and $|y, y\rangle$. Determine the density matrix for this ensemble using the linear polarization states of the photons as basis states.

Bonus: Find the reduced density matrix of particle 1 by itself. I recommend doing the “partial trace” on the density *operator* using a sum of bras and kets for particle 2, and only then turning the result into a density *matrix*.

We have the elements of the density matrix as

$$\begin{bmatrix} \langle x, x | \langle x, x | & \langle x, x | \langle y, y | \\ \langle y, y | \langle x, x | & \langle y, y | \langle y, y | \end{bmatrix}$$

Or is it

$\langle x, x \langle x, x $	$\langle x, x \langle y, y $	$\langle y, y \langle x, x $	$\langle y, y \langle y, y $
st			

Or do we have

$$\begin{bmatrix} |x\rangle\langle x||x\rangle\langle x| & |x\rangle\langle x||x\rangle\langle y| & |x\rangle\langle y||x\rangle\langle x| & |x\rangle\langle y||x\rangle\langle y| \\ etc & & & \end{bmatrix}$$

TODO

50/50 Mixes (5 pts) What is the density matrix (in the z basis) for a 50/50 mix of

(a) $|+z\rangle$ and $|-z\rangle$.

(a) $|+n\rangle$ and $|-n\rangle$.

Given that probabilities for measurement results, expectation values of observables, and time evolution are all completely determined by the density matrix, is there any way to tell whether someone has handed you a particle using procedure (a) or (b)?

a

We have $\frac{1}{2}$ of each state, so

$$\rho = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

b

We have the exact scenario in the $|n\rangle$ basis, since $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$

So it looks the same,

$$\rho = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

c

I don't think we would be able to tell, since we could measure in any basis and these two mixes look the exact same.

Maybe if time evolution were dependent on state polarization - like a magnetic field in some direction, but I don't think so since the states are identical.

$$\frac{d}{dt}\hat{\rho}(t) = \frac{1}{i\hbar}[\hat{H}(t), \rho(t)]$$

Townsend Problem 6.1 (5 pts)

(a) Use induction to show that $[\hat{x}^n, \hat{p}_x] = i\hbar n\hat{x}^{n-1}$. *Suggestion:* Take advantage of the commutation relation $[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$ in working out the commutators.

a

For $n = 1$, we have

$$[\hat{x}^1, \hat{p}_x] = i\hbar$$

For $n = 2$, we have

$$\begin{aligned} [\hat{x}^2, \hat{p}_x] &= \\ [xx^{n-1}, \hat{p}_x] &= \hat{x}[\hat{x}^{n-1}, \hat{p}_x] + [\hat{x}, \hat{p}_x]x^{n-1} \\ &= \hat{x}[\hat{x}^1, \hat{p}_x] + [\hat{x}, \hat{p}_x]x^1 \\ &= 2i\hbar x \end{aligned}$$

for $n = 3$

$$\begin{aligned}
[xx^{n-1}, \hat{p}_x] &= \hat{x}[\hat{x}^{n-1}, \hat{p}_x] + [\hat{x}, \hat{p}_x]x^{n-1} \\
&= \hat{x}[\hat{x}^2, \hat{p}_x] + [\hat{x}, \hat{p}_x]\hat{x}^2 \\
&= \hat{x}(2i\hbar\hat{x}) + i\hbar\hat{x}^2 \\
&= 3i\hbar\hat{x}^2
\end{aligned}$$

We can show that this result holds inductively

$$\begin{aligned}
[xx^{n-1}, \hat{p}_x] &= \hat{x}[\hat{x}^{n-1}, \hat{p}_x] + [\hat{x}, \hat{p}_x]x^{n-1} \\
&= \hat{x}(\hat{x}[\hat{x}^{n-2}, \hat{p}_x] + [\hat{x}, \hat{p}_x]x^{n-2}) + [\hat{x}, \hat{p}_x]x^{n-1} \\
&=
\end{aligned}$$

Where for $n = 1$, we get $2i\hbar x$ which propagates forwards.

For any n , we can plug in the result from the previous for the first term of the expansion, which will have the form

$$\hat{x}[\hat{x}^{n-1}, \hat{p}_x] = (n-1)i\hbar\hat{x}^{n-1}$$

as well as

$$[\hat{x}, \hat{p}_x]\hat{x}^{n-1} = i\hbar\hat{x}^{n-1}$$

The sum of these two is just

$$[\hat{x}^n, \hat{p}_x] = i\hbar nx^{n-1}$$

B

(b) Using the expansion

$$F(x) = F(0) + \left(\frac{dF}{dx}\right)_{x=0} x + \frac{1}{2!} \left(\frac{d^2F}{dx^2}\right)_{x=0} x^2 + \dots + \frac{1}{n!} \left(\frac{d^n F}{dx^n}\right)_{x=0} x^n + \dots$$

show that

$$[F(\hat{x}), \hat{p}_x] = i\hbar \frac{\partial F}{\partial x}(\hat{x})$$

TODO:

how do we know eigenvalues of $\hat{x}$ are being fed in?

C

(c) For the one-dimensional Hamiltonian

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + V(\hat{x})$$

show that

$$\frac{d\langle p_x \rangle}{dt} = \left\langle -\frac{dV}{dx} \right\rangle$$

We can rearrange the Hamiltonian, and have derivatives in space and derivatives in time with separation of variables. Those must be constant (because they aren't in terms of the other). We get

$$\frac{1}{2m} \frac{d\hat{p}_x^2}{dt} = -\frac{d\langle \hat{x} \rangle}{dx}$$

TODO

Townsend Problem 6.2 (5 pts)

Show that

$$\langle p|\hat{x}|\psi\rangle = i\hbar\frac{\partial}{\partial p}\langle p|\psi\rangle$$

and

$$\langle\varphi|\hat{x}|\psi\rangle = \int dp \langle p|\varphi\rangle^* i\hbar\frac{\partial}{\partial p}\langle p|\psi\rangle$$

What do these results suggest for how you should represent the position operator in momentum space?

Townsend 6.4 (5 pts)

(a) Show for a free particle of mass m initially in the state

$$\psi(x) = \langle x|\psi\rangle = \frac{1}{\sqrt{\sqrt{\pi}a}} e^{-x^2/2a^2}$$

that

$$\psi(x,t) = \langle x|\psi(t)\rangle = \frac{1}{\sqrt{\sqrt{\pi}[a + (i\hbar t/ma)]}} e^{-x^2/\{2a^2[1 + (i\hbar t/ma^2)]\}}$$

and therefore

$$\Delta x = \frac{a}{\sqrt{2}} \sqrt{1 + \left(\frac{\hbar t}{ma^2}\right)^2}$$

Suggestion: Start with (6.75) and take advantage of the Gaussian integral (D.7), but in momentum space instead of position space.

(b) What is Δp_x at time t ? *Suggestion:* Use the momentum-space wave function to evaluate Δp_x .

$$\langle x|\psi(t)\rangle = \int dp e^{-ip^2 t/2m\hbar} \langle x|p\rangle \langle p|\psi(0)\rangle$$

We can pull out constants and evaluate

$$\langle x|p\rangle = N e^{\frac{ipx}{\hbar}} = \sqrt{\frac{1}{2\hbar}} e^{\frac{ipx}{\hbar}}$$

$$\langle p|\psi\rangle = \int dx \langle p|x\rangle \langle x|\psi\rangle = \int dx \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar} \langle x|\psi\rangle \quad (6.57a)$$

$$\langle x|\psi\rangle = \int dp \langle x|p\rangle \langle p|\psi\rangle = \int dp \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \langle p|\psi\rangle \quad (6.57b)$$

$$\begin{aligned} \langle p|\psi(0)\rangle &= \psi(0) \\ &= \int \frac{1}{2\hbar} e^{\frac{-ipx}{\hbar}} \langle x|\psi(0)\rangle \end{aligned}$$

$$\langle x|\psi(t)\rangle = \frac{1}{\sqrt{\sqrt{a}}} \int dp \sqrt{\frac{1}{2\hbar}} e^{\frac{ipx}{\hbar}}$$

For reference:

For the Gaussian wave packet (6.62), $\langle p|\psi(0)\rangle$ is given by (6.68) and thus $\psi(x,t)$ is given by

$$\psi(x,t) = \langle x|\psi(t)\rangle = \int dp e^{-ip^2 t/2m\hbar} \langle x|p\rangle \langle p|\psi(0)\rangle \quad (6.75)$$

If you are proficient at carrying out Gaussian integrals (see Appendix D), it is straightforward to show that⁸

$$\psi(x,t) = \frac{1}{\sqrt{\sqrt{\pi}[a + (i\hbar t/ma)]}} e^{-x^2/\{2a^2[1 + (i\hbar t/ma^2)]\}} \quad (6.76)$$

Comparing $\psi^*(x,t)\psi(x,t)$ with its form (6.63) at $t=0$, we see that the position uncertainty is given by

$$\Delta x = \frac{a}{\sqrt{2}} \left(1 + \frac{\hbar^2 t^2}{m^2 a^4}\right)^{1/2} \quad (6.77)$$

Making the change of variables $x' = x - (b/2a)$, we find

$$I(a, b) = \int_{-\infty}^{\infty} dx e^{-ax^2+bx} = e^{b^2/4a} \int_{-\infty}^{\infty} dx' e^{-ax'^2} = e^{b^2/4a} \sqrt{\frac{\pi}{a}} \quad (\text{D.7})$$

B

TODO

Townsend 6.5 (5 pts)

Consider a wave packet defined by

$$\langle p|\psi\rangle = \begin{cases} 0 & p < -P/2 \\ N & -P/2 < p < P/2 \\ 0 & p > P/2 \end{cases}$$

- Determine a value for N such that $\langle\psi|\psi\rangle = 1$ using the momentum-space wave function directly.
- Determine $\langle x|\psi\rangle = \psi(x)$.
- Sketch $\langle p|\psi\rangle$ and $\langle x|\psi\rangle$. Use reasonable estimates of Δp_x from the form of $\langle p|\psi\rangle$ and Δx from the form of $\langle x|\psi\rangle$ to estimate the product $\Delta x \Delta p_x$. Check that your result is independent of the value of P . *Note:* Simply estimate rather than actually calculate the uncertainties.

A

B

TODO

Townsend 6.20 (5 pts)

Normalize the wave function

$$\langle x|\psi\rangle = \begin{cases} Ne^{-\kappa x} & x > 0 \\ Ne^{\kappa x} & x < 0 \end{cases}$$

Determine the probability that a measurement of the momentum p finds the momentum between p and $p + dp$ for this wave function. *Note:* This wave function is the energy eigenfunction for the delta function potential energy well of Problem 6.19.

Infinitely high barriers, infinitely thin well.